A Novel NMF Guided for Hyperspectral Unmixing From Incomplete and Noisy Data

Le Dong, Xiaoqiang Lu, *Senior Member, IEEE*, Ganchao Liu, and Yuan Yuan, *Senior Member, IEEE*

***Abstract*— The nonnegative matrix factorization (NMF)-combined spatial–spectral information has been widely applied in the unmixing of hyperspectral images (HSIs). However, how to select the appropriate similarity pixels and explore the spatial information and how to adapt the unmixing algorithm to complex data are both great challenges. In this article, we propose a novel unmixing method named spatial–spectral neighborhood preserving NMF (SSNPNMF) for incomplete and noisy HSI data. First, a spatial–spectral kernel regularizer is introduced to preprocess the HSI, which can reduce noise and complete missing elements. Second, a distance metric SSD based on spatial–spectral information is designed to select similar pixels in the image. Subsequently, the spatial–spectral relationship of the selected first *k* similar pixels is used to reconstruct the image and obtain the reconstruction matrix. Finally, the reconstruction matrix is used to constrain the abundances and improve the unmixing performance. Experimental results on synthetic data and Cuprite data indicate that SSNPNMF has a more effective unmixing performance compared with the state-of-the-art methods.**

***Index Terms*— Hyperspectral unmixing (HU), image reconstruction, nonnegative matrix factorization (NMF), spatial–spectral information.**

Notation and Definition

Y	Raw hyperspectral image
A	The endmember matrix
S	The abundance matrix
M	Number of bands
N	Number of pixels
P	Number of endmembers
$\mathcal{N}_k(y_i)$	k-nearest neighborhood of pixel y_i
(p_n, q_n)	Spatial coordinate of pixel y_n
r	the size of spatial window
y_i'	the pixel y_i after reconstruction
$\phi(y_i)$	Set of the neighborhood pixels set of y_i
$L_{Spa}(p, q)$	Spatial weight coefficient of p and q
$L_{Spe}(y_i, y_j)$	Spectral weight coefficient of y_i and y_j
I	Identity matrix
w_{ij}	Reconstruction weight of y_i and y_j
W	Reconstruction weight matrix

Manuscript received January 12, 2021; revised March 9, 2021 and July 6, 2021; accepted July 23, 2021. Date of publication August 30, 2021; date of current version January 26, 2022. This work was supported in part by the National Key Research and Development Project under Grant 2020YFB2103902, in part by the National Science Fund for Distinguished Young Scholars under Grant 61825603, in part by the Key Program of National Natural Science Foundation of China under Grant 61632018, and in part by the National Natural Science Foundation of China under Grant 62001390. *(Corresponding author: Yuan Yuan.)*

Le Dong, Ganchao Liu, and Yuan Yuan are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: y.yuan1.ieee@qq.com).

Xiaoqiang Lu is with Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shannxi 710119, China.

Digital Object Identifier 10.1109/TGRS.2021.3101504

I. Introduction

With the development of hyperspectral imaging technology, more and more detailed spectral information is captured by sensors, which provides strong support for ground observation [1], [2]. Hyperspectral image (HSI) is widely applied in areas such as surface feature classification [3], disaster detection [4], and environmental monitoring [5]. However, due to the limitations of low spatial resolution and complex feature distribution, the pixels in HSI are generally in a mixed state [6]–[8]. The purpose of hyperspectral unmixing (HU) is to extract all the pure materials (i.e., endmembers) and their corresponding fraction (i.e., abundances) from each pixel [9].

HU is a very challenging problem. An effective unmixing algorithm must rely on the establishment of a proper mixing model, accurate description of the abundance matrix characteristics, tolerance to the complex ground environment, and uncertain noise in the image, as well as an effective optimization method. However, most of the existing unmixing methods do not perform well on these factors, which ignores or fails to handle.

First, HSI mixing models are mainly divided into two categories: linear mixing models (LMMs) and non-LMMs [10]. The LMM assumes that each pixel is a linear combination of endmember, while the nonlinear model assumes that each endmember undergoes multiple interactions before reaching the sensor [11]. The imaging principle of the LMM is simple, and it is easy to model and implement. Meanwhile, the nonlinear mixing hypothesis is more complicated in imaging mechanisms, and it is difficult to realistically simulate. Therefore, a lot of studies on HU algorithms are based on LMM, including geometry-based methods, sparse regression-based methods, and statistics-based methods [12], [13]. Due to the huge advantage of nonnegative matrix factorization (NMF) in finding the basis of data and the coefficients of data under this basis [14], [15], this article builds an efficient unmixing algorithm based on NMF.

Second, the abundance matrix has many characteristics that are worth mining to improve the unmixing performance of the NMF framework. As we all know, HSI contains rich spectral information and spatial context information. In recent years, plenty of works have been devoted to exploring the spatial

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

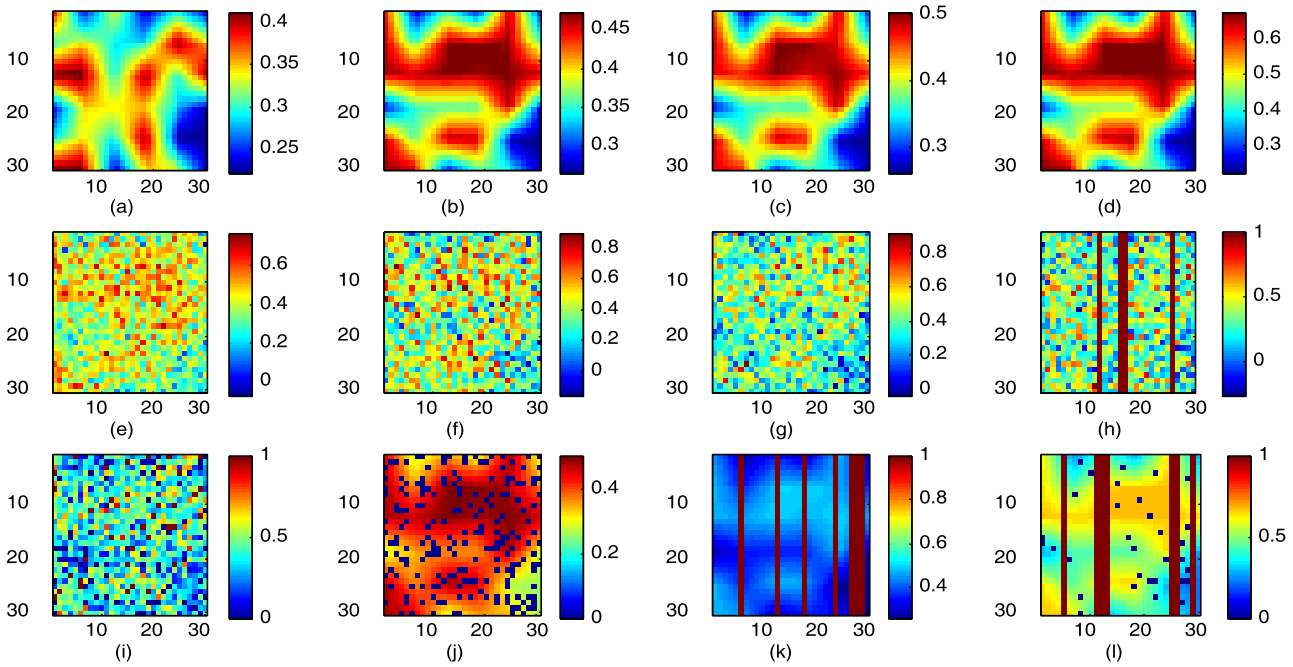


Fig. 1. Example bands of synthetic data. (a) Original band 7. (b) Original band 13. (c) Original band 14. (d) Original band 23. (e) Gaussian noise band 14 of Case 1. (f) Gaussian noise band 14 of Case 2. (g) Impulse noise band 14 of Case 3. (h) Dead line and mixed noise band 7 of Case 4. (i) Mixed noise band 13 of Case 4. (j) Dead pixels band 14 of Case 5. (k) Deadlines band 14 of Case 6. (l) Deadlines and dead pixels band 23 of Case 6.

relationship of pixels in images, among which the most direct method is to determine the similarity of pixels.

Some works consider each pixel as an individual and place each pixel in a Laplacian graph, in which the adjacency relationship of points can describe whether the pixels are similar or not in the whole image. For example, a manifold constraint aiming at exploring the latent spatial relations is proposed in [16]. Tong *et al.* [17] propose an adaptive graph to regularize multilayer NMF, which can automatically determine the best neighbor points of pixels and assign corresponding weights. However, these methods rely on the establishment of graphs, and the computational complexity will be very high for large images.

Some works assume that ground features have local smoothness in space, and consider that adjacent pixels have similar abundances. They usually use image segmentation or clustering to obtain spatial groups. Pixels in each spatial group share similar abundances, which can reflect the continuous distribution characteristics in geographic space. For instance, Iordache *et al.* [18] introduce a total variational regularizer term for imposing the spatial consistency of pixels. Wang *et al.* [19] integrate the group structure information of HSI into NMF to improve the stability of the result, where each superpixel is a spatial group that shares the same sparse structure. Unfortunately, these methods are more affected by noise, especially for images with missing entries, and it is difficult to correctly evaluate the neighborhood relationship, which will greatly affect the unmixing performance.

Besides, sparsity is an important property of the abundance. Many researchers have focused on the study of sparse regularizer and embedded it in NMF to improve the stability of HU recently. As we all know, L_0-norm is outstanding idea to express sparsity [20], but it is an NP-hard problem. Therefore, L_1-norm regularization is proposed as a relaxation of the solution of L_0-norm in [21]. References [22]–[24] introduce $L_{1/2}$-norm, which shows more effective than L_1-norm in impulse sparseness. Furthermore, L_p-norm ($0 < p < 1$) [25] is also introduced to increase the sparsity of abundance, representing an important method for expressing sparsity.

Third, HSI is susceptible to noise interference during the acquisition process. As shown in Fig. 1, different types of noise interference, dead pixels, and deadlines are widely present in HSI. The presence of noise can reduce the performance of unmixing algorithms; therefore, researchers have explored unmixing algorithms that can suppress noise. For instance, [26] points out that there is not only Gaussian noise in HSI but also sparse noise, in which sparse noise and Gaussian noise are both modeled to constrain the abundance maps. He *et al.* [27] propose a weighted sparse regularizer to reduce the noise of the algorithm sensitivity. In [28], an antinoise model is specially developed for unmixing, which considers the additive, multiplicative, and mixed noises. Nevertheless, the existing unmixing algorithms rarely consider the situation of missing entries and contaminated parts of the image. They usually regard the same level of noise level on each band of the image, which greatly affects the understanding of the unmixing performance. How to correctly deal with these missing entries in the HU and restore them to the correct values is an important issue that cannot be ignored.

In view of the above three points, designing a strong, unified framework with superior performance can still accurately extract the spatial information contained during the presence of missing elements and large-area noise in the image. Proper

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

constraints on the abundance matrix are essential to the task of unmixing HSIs. Unfortunately, most of the existing algorithms cannot meet the above three points at the same time; especially in the absence of data, most of the strategies describing spatial relationships will be ineffective. Therefore, here, we take the LMM as the basic assumption, put the unmixing problem in the NMF framework, and find a novel method that can best fit the HSI spatial information to improve the robustness and adaptability.

In this article, a novel NMF based on spatial–spectral neighborhood preserving method is designed, which is used to perform mixed pixels decomposition. In addition, we also introduce a new distance measurement to measure the similarity between pixels to effectively discover the spatial information when there are missing entries or the image is greatly disturbed by noise. It is worth noting that, when the image is complete, the algorithm design process still holds. The main contributions involved in this article can be summarized as follows.

1) A framework called spatial–spectral neighborhood preserving NMF (SSNPNMF) is proposed to perform the unmixing from incomplete and noisy data.
2) A spatial–spectral kernel regularization is introduced to preprocess the image, which can smooth the HSI and reduce noise interference.
3) A new distance metric named spatial–spectral joint distance (SSD) is designed to measure the similarity between pixels. As a result, not only can it make up for the shortcomings of the traditional distance measurement method when processing data with missing elements but it can also use the spatial and spectral information in the image to the greatest extent possible.
4) The spatial–spectral information of the first k similar pixels is used to reconstruct the image, and the obtained reconstruction matrix is applied to constrain the abundance. At the same time, it has been proven to be effective on synthetic and real data.

The remainder of this article is described in detail as follows. Section II introduces the key related work. Our proposed SSNPNMF algorithm and specific optimization process are presented in Section III. Section IV describes the experimental results on synthetic. Section V details the unmixing results of real data. The summary of this article is introduced in Section VI.

II. Related Works

In this section, we show some basic concepts, including LMM based on the NMF framework and locally linear embedding (LLE). For easy understanding, all the mathematical notations used in this article are summarized in Nomenclature.

A. Linear Mixing Model

LMM assumes that the pixels in HSI are all linear combinations of pure endmembers. Let $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{M\times N}$ represent hyperspectral data with L bands and N pixels. Then, LMM can be described as

$$Y = \mathrm{AS} + E \tag{1}$$

where $A = [a_1, a_2, \ldots, a_P] \in \mathbb{R}^{M\times P}$ is endmember matrix with P pure materials, $S = [s_1, s_2, \ldots, s_N] \in \mathbb{R}^{P\times N}$ denotes the abundance matrix, and $E \in \mathbb{R}^{M\times N}$ represents the additional noise in the image. Under LMM, there are two constraints on each abundance vector $s_{n\in[1,N]}$ that should be observed: *abundance nonnegativity constraint* (ANC) and *abundance sum-to-one constraint* (ASC). ANC is to ensure that each item has practical significance, while ASC indicates that the sum of the coding coefficients of the endmembers in each pixel y_n should be 1. Specifically, they can be formulated as

$$s_{\mathrm{in}} \geq 0, \quad i = 1, 2, \ldots, P, \quad \sum_{i=1}^{P} s_{\mathrm{in}} = 1 \tag{2}$$

where s_{in} represents the proportion of the ith endmember in the nth pixel.

NMF is widely used to solve the endmember and abundance matrices described in (1). The corresponding optimization problem can be given as follows[1]:

$$\min_{A,S\geq 0} ||Y - \mathrm{AS}||_F^2. \tag{3}$$

Unfortunately, the above problem is an ill-posed problem [16], so it is important to add proper regularization and constraint terms. In recent years, many regularization models have been proposed, such as sparsity-based NMF models [29], graph-based models [30], local spatial information-based models [31], and nonlocal spatial information-based models [32]. Their optimization problem can be expressed as

$$\begin{aligned} &\min_{A,S} ||Y - \mathrm{AS}||_F^2 + \lambda J(S) \\ &\text{s.t. } s_{i,n} \geq 0, \quad \sum_{i=1}^{P} s_{i,n} = 1 \end{aligned} \tag{4}$$

where $J(S)$ denotes the constraint item, which generally represents some prior characteristics of pixels or abundance vectors, and λ is the weight parameter used to balance the regularization and the original NMF term.

B. Locally Linear Embedding

The LLE assumption means that each data point can be linearly reconstructed by its neighborhood in the original space; then, it can be linearly expressed from its neighbors using the same coefficients in a low-dimensional subspace [33]. Its purpose is to maintain the original local manifold structure no matter how the data points are transformed.

Specifically, $\mathcal{N}_k(y_n)$ denotes the k-neighbors of point y_n. First, the k-nearest neighborhood of pixel is used to reconstruct each data point. The reconstruction coefficients can be calculated by minimizing the following functions:

$$\begin{aligned} &\min \left\| y_n - \sum_{y_j\in\mathcal{N}_k(y_n)} W_{jn} y_j \right\|^2 \\ &\text{s.t.} \sum_{y_j\in\mathcal{N}_k(y_n)} W_{jn} = 1 \end{aligned} \tag{5}$$

[1]In this article, unless otherwise specified, all matrices and vectors satisfy nonnegative constraints.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Construct the Reconstruction Matrix W

Input: The raw hyperspectral data $\mathrm{Y} = [\mathrm{y}_1, \mathrm{y}_2, \ldots, \mathrm{y}_n] \in \mathbb{R}^{M \times N}$, the size of window r, the neighbor number $K > 0$.
Output: The similarity matrix W.

for $i = 1; i \leq N$ **do**
  Find all pixels in the $r \times r$ neighborhoods of y_i by applying the rule (10);
  for $j = 1; j \leq r^2$ **do**
    Calculate the spatial weights L_{Spa} and spectral weights L_{Spe} via (12);
    $j \leftarrow j + 1$;
  Compute the pixels $\mathrm{y}'_i = \frac{\sum_{j \in \phi(\mathrm{y}_i)} L_{\mathrm{Spa}}(p,q) L_{\mathrm{Spe}}(\mathrm{y}_i, \mathrm{y}_j) \mathrm{y}_j}{\sum_{j \in \phi(\mathrm{y}_i)} L_{\mathrm{Spa}}(p,q) L_{\mathrm{Spe}}(\mathrm{y}_i, \mathrm{y}_j)}$;
  $i \leftarrow i + 1$;
for $i = 1; i \leq N$ **do**
  Calculate the distance between pixels by (14);
  Find the K nearest neighbors $\mathrm{y}_i^1, \mathrm{y}_i^2, \ldots, \mathrm{y}_i^K$;
  Calculate the reconstruction matrix $\mathbf{W} = [\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_K]$ for each pixel via (16);
  $i \leftarrow i + 1$;
Return $\mathrm{W} = [\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_N]$;

where W_{jn} is the reconstruction weight between y_n and y_j, and $W_{jn} = 0$ if $y_j \notin \mathcal{N}_k(y_n)$.

Second, since W mentioned above describes the local geometric structure among data, s_n in low-dimensional embedding of y_n can be approximately linearly reconstructed with the same weight W and k-neighbors according to LLE. The corresponding construction function in the low-dimensional subspace is

$$\begin{aligned} &\min \sum_{i=1}^{N} \left\| s_n - \sum_{s_j \in N_k(\mathrm{s}_n)} W_{jn} s_j \right\|^2 \\ &= \mathrm{tr}\big(S(I-W)(I-W)^T \mathrm{S}^T\big) \\ &= \mathrm{tr}\big(\mathrm{SLS}^T\big) \end{aligned} \tag{6}$$

where $L = (I - W)(I - W)^T$ and $I \in \mathbb{R}^{N \times N}$ is an identity matrix.

III. Proposed Method

When the sensor collects surface information, HSI would inevitably suffer from noise or cause dead pixels, deadlines, or unknown elements, as shown in Fig. 1. Therefore, we mainly consider the HU from incomplete and noise data. Due to the objective existence of low spatial resolution and assumptions based on LMM, the HSI is low rank and can be decomposed into two nonnegative factors: one of which represents the endmember matrix and the other is the abundance matrix.

The rich information of the spatial–spectral in HSI is mainly considered to guide the unmixing, which can supplement information to make up for the loss of disturbance and missing in the data. Therefore, in this section, we propose a novel framework named SSNPNMF to perform HU from incomplete and noise data. The specific steps are introduced in detail in the following.

Given hyperspectral data $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{M \times N}$ with missing entries, each pixel can be expressed as a vector $y_n \in \mathbb{R}^M, n = (1, 2, \ldots, N)$. Therefore, SSNPNMF can be described as follows:

$$\begin{aligned} &\min_{\mathrm{A,S,Y} \geq 0} \ \tfrac{1}{2} ||\mathrm{AS} - Y||_F^2 + \lambda_1 J(S) \\ &\text{s.t. } s_{i,n} \geq 0, \ \sum_{i=1}^{P} s_{i,n} = 1 \end{aligned} \tag{7}$$

where $J(S)$ denotes the spatial–spectral constraint of abundance S and λ_1 is a parameter to balance the weight of $J(\mathrm{S})$.

In addition, we focus on the sparsity of the abundance matrix S. According to [34]–[36], $L_{1/2}$ regularization can achieve superior results in constraining the sparsity of the abundance coefficients. Therefore, here, $L_{1/2}$ is used to sparse the coefficient vector. Specifically,

$$\begin{aligned} &\min_{\mathrm{A,S}} \|\mathrm{Y} - \mathrm{AS}\|_{\mathrm{F}}^2 + \lambda_1 J(S) + \lambda_2 \|S\|_{\frac{1}{2}} \\ &\text{s.t. } s_{i,n} \geq 0, \ \sum_{i=1}^{P} s_{i,n} = 1 \end{aligned} \tag{8}$$

where λ_2 is used as a weighting coefficient to adjust the sparsity of the abundance matrix, and $||\mathrm{S}||_{(1/2)}$ can be defined as

$$||S||_{\frac{1}{2}} = \sum_{i=1}^{P} \sum_{j=1}^{N} S_{i,j}^{\frac{1}{2}}. \tag{9}$$

A. Similarity Matrix W

As mentioned in the previous discussion, it can be concluded that the spatial information among pixels is of great significance for HU. However, for images with incomplete and noisy data, using existing similarity measures would cause large errors. The conventional similarity calculation method is generally based on the Euclidean distance between pairwise pixels, which is easily affected by data disturbance [37]. Especially when there is incomplete data or noise interference in the image, its judgment will lose accuracy. Therefore, inspired by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

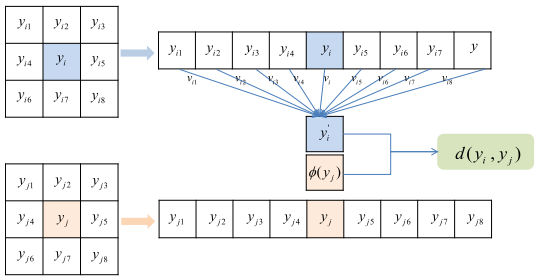


Fig. 2. Illustration of SSD metric, where y_i and y_j denotes pixels, $\phi(y_j)$ is the neighbor pixels set of y_j, y_i' represents the pixels y_i after reconstruction by neighbor pixels, and $d(y_i, y_j)$ is the similarity distance of y_i and y_j.

Huang *et al.* [38], we propose a novel similarity measurement named SSD that combines spatial–spectral information in this article.

First, a spatial–spectral kernel regularization is introduced to preprocess the HSI. Specifically, assuming that the spatial coordinate of current pixel $y_n \in \mathbb{R}^M$ is (p, q), then the neighborhood pixels with y_n as the center coordinate can be expressed as

$$\phi(y_n) = \left\{ y_n(p,q) | \begin{array}{l} p \in [p_n - t, p_n + t] \\ q \in [q_n - t, q_n + t] \end{array} \right\} \tag{10}$$

where $t = (r-1)/2$ and r represents the size of the spatial window. Generally speaking, it is a positive odd number. As shown in Fig. 2, $r = 3$, and $\phi(y_n)$ denotes a set of eight neighborhood pixels of y_n. Encouraged by the principle and advantages of bilateral filtering [39], here, we consider both the spatial domain and the pixel range domain, so we can get the pixels after pretreatment as follows:

$$y_n' = \frac{\sum_{y_j \in \phi(y_n)} L_{\text{Spa}}(p,q) L_{\text{Spe}}(y_n, y_j) y_n}{\sum_{y_j \in \phi(y_n)} L_{\text{Spa}}(p,q) L_{\text{Spe}}(y_n, y_j)} \tag{11}$$

where y_j denotes the neighborhood pixel y_n in $\phi(y_n)$, $L_{\text{Spa}}(p,q)$ is the weight coefficient of p and q in the spatial domain, and $L_{\text{Spe}}(y_n, y_j)$ represents the weight coefficient of y_n and y_j in the spectral domain. Mathematically, spectral and spatial weights are calculated as follows, respectively:

$$\begin{aligned} L_{\text{Spa}}(n, j) &= \exp\left\{ \frac{-||n-j||^2}{\sigma_{\text{spa}}} \right\} \\ L_{\text{Spe}}(y_n, y_j) &= \exp\left\{ \frac{-||y_n - y_j||^2}{\sigma_{\text{spe}}} \right\} \end{aligned} \tag{12}$$

where σ_{Spa} and σ_{Spe} are used to control the contribution of the distance in the spatial domain and the spectral value in the spectral domain to the current pixel when the pixels are reconstructed, respectively. The benefit of using the above spatial–spectral method to reconstruct data is very obvious. For each pixel, synthesizing information in the spatial and spectral domains simultaneously can effectively suppress noise in the image and keep the homogeneous regions smooth.

Subsequently, we establish a new distance metric SSD to explore the similarity between the data, as shown in Fig. 2. For pixels y_i and y_j in HSI, the similarity of them can be described as $d(y_i, y_j)$. It can be calculated as follows:

$$d(y_i, y_j) = d(y_i', \phi(y_j)) \tag{13}$$

Algorithm 2 Overall Algorithm of SSNPNMF for Unmixing

Input: The raw data $\mathrm{Y} \in \mathbb{R}^{M \times N}$, the number of endmembers P, the size of window r, and the parameters $\lambda_1 > 0$, $\lambda_2 > 0$.
Output: A, S, and Y.
Initialize $\mathrm{A} \in \mathbb{R}^{M \times P}$ by VCA-based method, $\mathrm{S} \in \mathbb{R}^{P \times N}$ by FLCL, Y=AS.
while *not convergence* **do**
  Update endmember A by adopting the (29);
  Update abundance S via applying the (30);
Return Endmeber A, Abundance S.

where y_i' is the pixel y_i after reconstruction described above, and $\phi(y_j)$ denotes the neighbor space pixel set of pixel y_j. In detail, $d(y_i, y_j)$ can be defined as

$$d(y_i, y_j) = \frac{\sum_{y_n \in \phi(y_j)} v_{\text{in}} \left\| y_i' - y_n \right\|}{\sum_{y_n \in \phi(y_j)} v_{\text{in}}} \tag{14}$$

in which v_{in} describes the degree of similarity between y_i' and y_n in the neighbor space pixel set $\phi(y_j)$ of y_j. It can be obtained by the heat kernel function as

$$v_{\text{in}} = \exp\left\{ \frac{-\left\| y_i' - y_n \right\|^2}{\sigma^2} \right\} \tag{15}$$

where $\sigma = \sum_{y_n \in \phi(y_j)} ||y_i' - y_n||^2$ is a smooth kernel.

According to the previous definition of the pixel similarity, for each pixel y_i in an HSI, we can find the K pixels that are most similar to it in the global area of the image. Finally, the K nearest neighbors of each pixel are used to construct a global similarity matrix $W \in \mathbb{R}^{K \times N}$. Generally speaking, the closer spatial positions of the pixels in the HSI, the higher their similarity. Therefore, here, we also add the spatial coordinate relationship of the pixels to calculate the reconstruction coefficient. Let $h_i^k = y_i - y_i^k / ((p_i - p_i^k)^2 + (q_i - q_i^k)^2)^{1/2}$, $k = 1, 2, \ldots, K$; the specific calculation method of the reconstruction coefficient is

$$\begin{aligned} \Psi(W) &= \min \sum_{i=1}^{N} \left\| \frac{1}{\sqrt{(p_i - p_j)^2 + (q_i - q_j)^2}} \left(y_i - \sum_{j=1}^{K} W_{ji} y_i^j \right) \right\|^2 \\ &= \min \sum_{i=1}^{N} \left\| \sum_{j=1}^{K} \left(\frac{y_i - y_i^j}{\sqrt{(p_i - p_j)^2 + (q_i - q_j)^2}} \right) W_{ji} \right\|^2 \\ &= \min \sum_{i=1}^{N} W_i^T D_i W_i \\ &\quad \text{s.t.} \sum_{y_j \in N_K(y_i)} W_{ji} = 1 \end{aligned} \tag{16}$$

where $W_i = [W_{i1}, W_{i2}, \ldots, W_{iK}]^T \in \mathbb{R}^{K \times 1}$ is a column vector, W_{ji} denotes the jth row of W_i, y_i^j $(j = 1, 2, \ldots, K)$ represents the jth nearest neighbor of y_i, and $D_i = [h_i^1, h_i^2, \ldots, h_i^K]^T [h_i^1, h_i^2, \ldots, h_i^K]$. According to [40],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

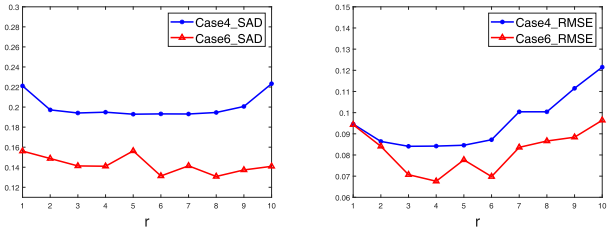


Fig. 3. Performance of SSNPNMF with respect to parameter r on Case 4 and Case 6 data. (a) SAD. (b) RMSE.

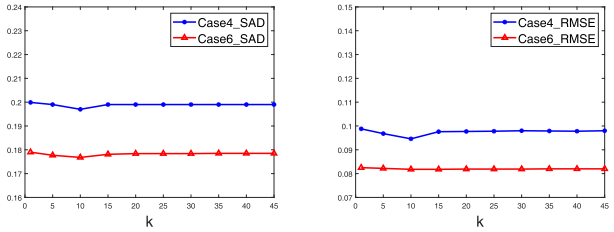


Fig. 4. Performance of SSNPNMF with respect to parameter k on Case 4 and Case 6 data. (a) SAD. (b) RMSE.

the solution of (16) can be obtained as

$$W_{ij} = \frac{\sum_{m=1}^{K} \left(z_i^{jm}\right)^{-1}}{\sum_{p=1}^{K}\sum_{q=1}^{K} \left(z_i^{pq}\right)^{-1}} \tag{17}$$

where $z_i^{jm} = (h_i^j)^T h_i^m$ and $z_i^{pq} = (h_i^p)^T h_i^q$.

To sum up, the similarity matrix W makes full use of spatial–spectral information contained in HSIs. The detailed calculation process is summarized in Algorithm 1. Then, the similarity relationship between pixels is transferred to the abundance. For the ith pixel, the reconstruction relationship of the abundance vector s_i can be expressed as

$$\begin{aligned} &\min \sum_{i=1}^{N} ||s_i - \sum_{s_j \in \mathcal{N}_K(s_i)} W_{ij} s_j||^2 \\ &= \mathrm{tr}\big(S(I-W)(I-W)^T S^T\big) \\ &= \mathrm{tr}\big(\mathrm{SLS}^T\big) \end{aligned} \tag{18}$$

in which I is an identity matrix and $L = (I - W)(I - W)^T$.

B. SSNPNMF Model

According to Cai *et al.* [41], the data correlation in the original space can be well preserved in the learned low-dimensional space by minimizing the graph Laplacian regularizer. Therefore, in this article, we can get the objective function as follows:

$$\begin{aligned} &\min_{A,S,Y\geq 0} \ \tfrac{1}{2}\|\mathrm{AS} - Y\|_F^2 + \lambda_1 \mathrm{Tr}\big(\mathrm{SLS}^T\big) + \lambda_2 \|S\|_{\frac{1}{2}} \\ &\text{s.t.} \ \ s_{i,n} \geq 0, \ \sum_{i=1}^{P} s_{i,n} = 1 \end{aligned} \tag{19}$$

where s_{in} represents the proportion of i endmember in the nth pixel. λ_1 is used to control the weight of the Laplacian regularizer, and λ_2 is a parameter used to adjust the sparsity of abundance. Here, we called (19) SSNPNMF.

C. Optimization

It can be seen that, although (19) is nonconvex, the subproblems for each variable are solvable. In this section, the optimization will be described in detail.

Similar to Lu *et al.* [16], to solve (19), the nonnegative constraints $A \geq 0$ and $S \geq 0$ are considered first. Let its Lagrange multipliers be $\Phi^{M\times P} \geq 0$ and $\Psi^{P\times N} \geq 0$. Since $\|Y\|_F = (\mathrm{Tr}(A^T A))^{1/2}$, the Lagrange function $\mathcal{L}$ can be defined as

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}\|Y - \mathrm{AS}\|_F^2 + \lambda_1 \mathrm{Tr}\big(\mathrm{SLS}^T\big) + \lambda_2 \|S\|_{\frac{1}{2}} \\ &\quad -\mathrm{Tr}\big(\Phi A^T\big) - \mathrm{Tr}\big(\Psi S^T\big) \\ &= \frac{1}{2}\mathrm{Tr}\big(YY^T\big) - \mathrm{Tr}\big(YS^T A^T\big) + \frac{1}{2}\mathrm{Tr}\big(\mathrm{AS}S^T A^T\big) \\ &\quad +\lambda_1 \mathrm{Tr}\big(\mathrm{SLS}^T\big) + \lambda_2 \|S\|_{\frac{1}{2}} - \mathrm{Tr}\big(\Phi A^T\big) - \mathrm{Tr}\big(\Psi S^T\big). \end{aligned} \tag{20}$$

Then, let the partial derivative of $\mathcal{L}$ with respect to A and S be 0; we can get

$$-YS^T + \mathrm{AS}S^T = \Phi \tag{21}$$

$$-A^T S + A^T \mathrm{AS} + 2\lambda_1 \mathrm{SL} + \frac{1}{2}\lambda_2 S^{-\frac{1}{2}} = \Psi. \tag{22}$$

The *Karush–Kuhn–Tucker* (KKT) conditions are $\Phi_{\mathrm{mp}} A_{\mathrm{mp}} = 0$ and $\Psi_{\mathrm{pn}} S_{\mathrm{pn}} = 0$; it can be concluded that

$$\big(-YS^T + \mathrm{AS}S^T\big).*A = 0 \tag{23}$$

$$\left(-A^T S + A^T \mathrm{AS} + 2\lambda_1 \mathrm{SL} + \frac{1}{2}\lambda_2 S^{-\frac{1}{2}}\right).*S = 0 \tag{24}$$

where .* denotes the elementwise multiplication between matrices. Then, we do a split on L

$$L = L^+ - L^- \tag{25}$$

where

$$L_{ij}^+ = \frac{|L_{ij}| + L_{ij}}{2}, \quad L_{ij}^- = \frac{|L_{ij}| - L_{ij}}{2}. \tag{26}$$

Therefore, (22) can be rewritten as

$$\begin{aligned} \big(-YS^T + \mathrm{AS}S^T\big).*A = 0 \\ \left(-A^T Y + A^T \mathrm{AS} + 2\lambda_1 \mathrm{SL}^+ - 2\lambda_1 \mathrm{SL}^- + \frac{1}{2}\lambda_2 S^{-\frac{1}{2}}\right).*S = 0. \end{aligned} \tag{27}$$

The following update rule can be obtained as:

$$A \leftarrow A.*\sqrt{\frac{YS^T}{\mathrm{AS}S^T}} \tag{28}$$

$$S \leftarrow S.*\sqrt{\frac{A^T Y + 2\lambda_1 SL^-}{A^T \mathrm{AS} + 2\lambda_1 SL^+ + \frac{1}{2}\lambda_2 S^{-\frac{1}{2}}}}. \tag{29}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

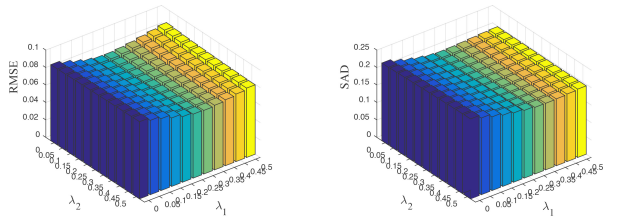


Fig. 5. RMSE and SAD versus the parameters λ_1 and λ_2 of the proposed method on Case 4 data.

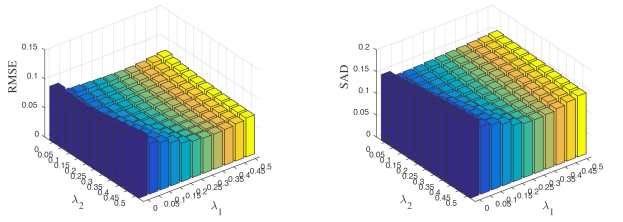


Fig. 6. RMSE and SAD versus the parameters λ_1 and λ_2 of the proposed method on Case 6 data.

The details in the preceding statement can help us understand how the SSNPNMF algorithm optimizes and updates each matrix. Now, combining these steps, we can get the complete algorithm of SSNPNMF, as shown in Algorithm 2.

D. Implementation Issues

In this section, we add some details of the algorithm SSNPNMF. The first is how the algorithm satisfies the ASC constraint. Here, we adopt a method similar to [42]. When using the update rule (29), we first transform the data and the endmember matrix accordingly and replace Y and A with $\tilde{Y}$ and $\tilde{A}$, respectively. Specifically,

$$\tilde{Y} = \begin{bmatrix} Y \\ \delta 1_N^T \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} A \\ \delta 1_P^T \end{bmatrix} \tag{30}$$

where δ is a parameter used to adjust the ASC. In this article, set δ to 15. Another detail is how to stop the algorithm. Here, we set an error tolerance threshold. When the reconstruction error is below this threshold for ten consecutive times, the algorithm stops iterating.

IV. Experiments With Synthetic Data

In this section, to evaluate the effectiveness of the proposed algorithm SSNPNMF, we first conduct experiments on six synthetic datasets. To ensure the fairness of experiments, we run 20 times for each experiment and take the average and standard deviation of the evaluation metrics as the final unmixing results.

A. Data

First, we construct basic data. The four spectra regarded as pure endmembers are randomly selected from the USGS spectral library [43] for fitting data. Specifically, we use a method similar to [16] to synthesize the basic hyperspectral data, where the size of the image is 30×30, and the number of spectral bands is 188. It is worth noting that the hyperspectral data do not contain pure pixels, that is, the purity is set to less than 1. The four bands of this original data are shown in Fig. 1(a)–(d).

Subsequently, to solve the problems raised in our article, we use this data as a benchmark and add six different interferences to verify the unmixing effect of our method when there are a large number of defects in hyperspectral data. The interferences mainly include Gaussian noise, impulse noise, dead pixel, and dead line. Among them, according to Cao *et al.* [44], the added impulse noise refers to salt and pepper noise, which is typical impulse noise, specifically, as follows.

Case 1: The zero-mean Gaussian noise with a variance of 0.1 is added to each band of the original data. The 14th band of Case 1 is shown in Fig. 1(e).

Case 2: The zero-mean Gaussian noise is randomly added to 20 bands of the original data, where the variance is set to a random variable between 0.1 and 0.2. The 14th band of Case 2 is shown in Fig. 1(f).

Case 3: Add the Gaussian noise with a variance of 0.1–0.2 to each band of the original data (similar to Case 2), and randomly select 20 bands then add impulse noise to them, where the percentage of impulse noise is 10%. The 14th band of Case 3 is shown in Fig. 1(g).

Case 4: Add the similar interference described in Case 3, including the Gaussian noise and the impulse noise, to each band of the HSI. In addition, randomly select 20 bands to add deadlines with a width of 1–3. The seventh and 13th bands of Case 4 are shown in Fig. 1(h) and (i).

Case 5: Black occasion and missing pixels are randomly added to the original data, and the missing amount accounts for 20% of the original pixels. The 14th band of Case 5 is shown in Fig. 1(j).

Case 6: Based on Case 5, randomly select 20 bands, add deadlines to them, and the width is randomly 1–3. The 14th and 23th bands of Case 4 are shown in Fig. 1(k) and (l).

B. Evaluation Metrics

Two very popular unmixing effect evaluation indicators are used to evaluate the performance of the all algorithms: *root-mean-square error* (RMSE) and *spectral angle distance* (SAD). For the ith endmember, RMSE is used to calculate the error between the reference abundance vector $\hat{S}_i$ and the estimated abundance S_i, while SAD is used to evaluate the distance between the estimated endmember A_i and the ground truth of endmember $\hat{A}_i$. They are defined as follows, respectively:

$$\text{RMSE} = \left(\frac{1}{n} \left| S_i - \hat{S}_i \right|^2 \right)^{\frac{1}{2}} \tag{31}$$

$$\text{SAD} = \cos^{-1} \left(\frac{\hat{A}_i^T A_i}{\left\| \hat{A}_i \right\| \left\| A_i \right\|} \right). \tag{32}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS ON CASE 1–6 SYNTHETIC DATA

Case	1		2		3	
Metrics	RMSE	SAD	RMSE	SAD	RMSE	SAD
$L_{1/2}$-RNMF	0.0412±0.0024	0.0479±0.0071	0.0633±0.0124	0.0807±0.0145	0.0709±0.0067	0.0875±0.0040
CSNMF	0.0494±0.0096	0.0689±0.0087	0.0665±0.0228	0.0802±0.0095	0.0717±0.0123	0.0925±0.0071
RCMF	0.0738±0.0384	0.0974±0.0556	0.0922±0.0672	0.1329±0.0920	0.0961±0.0645	0.1512±0.0764
NMF-QMV	0.0517	0.1469	0.0720	0.1769	0.0811	0.1806
SGSNMF	0.0767±0.0026	0.1559±0.0037	0.0784±0.0204	0.1722±0.0152	0.0850±0.0082	0.1825±0.0018
GLNMF	0.0423±0.0013	0.0504±0.0039	0.0625±0.0114	0.0762±0.0095	0.0638±0.0038	0.0833±0.0035
SSPNMF	**0.0394±0.0016**	**0.0401±0.0067**	**0.0564±0.0109**	**0.0562±0.0024**	**0.0582±0.0054**	**0.0818±0.0165**
Case	4		5		6	
Metrics	RMSE	SAD	RMSE	SAD	RMSE	SAD
$L_{1/2}$-RNMF	0.0928±0.0084	0.2156±0.0040	0.0991±0.0148	0.1021±0.0065	0.1066±0.0211	0.1765±0.0240
CSNMF	0.0885±0.0126	0.2076±0.0290	0.0952±0.0192	0.1199±0.0167	0.1182±0.0208	0.1970±0.0210
RCMF	0.1164±0.0632	0.2246±0.0657	0.1301±0.0141	0.1626±0.0088	0.1011±0.0896	0.1372±0.0834
NMF-QMV	0.1124	0.3578	0.1166	0.2015	0.1469	0.2049
SGSNMF	0.1153±0.0138	0.2973±0.0012	0.0949±0.0067	0.2057±0.0020	0.1133±0.0077	0.2217±0.0094
GLNMF	0.0946±0.0205	0.2029±0.0066	0.0967±0.0181	0.0960±0.0217	0.1040±0.0196	0.1683±0.0213
SSPNMF	**0.0750+0.0109**	**0.1879+0.0113**	**0.0623+0.0035**	**0.0916+0.0271**	**0.0709+0.0024**	**0.1357+0.0130**

C. Benchmarking

In this article, we mainly compare the effectiveness of our algorithm through the current popular NMF-based method. Due to the sensitivity of NMF to initial values, we first initialize endmember and abundance matrix through the VCA-FCLS[2] [45] method to ensure that all algorithms are in a relatively fair comparison. Since our method also uses the $L_{1/2}$ sparse constraint, we compare the NMF unmixing method RNMF under the $L_{1/2}$ constraint [26]. Its purpose is to illustrate the effectiveness of our proposed abundance constraint. At present, the NMF-based unmixing methods are mostly looking for the method that can more accurately explore the spatial structure of HSIs so that they can be fused with spectral information to perform high-precision unmixing in the form of space-spectrum joint. Therefore, this article compares several popular methods, including CSNMF [46], SGSNMF[3] [19], and NMF-QMV[4] [47]. Since our algorithm is mainly for HSIs disturbed by noise or dead pixels, we also compared the RCMF[5] algorithm, which was also specifically proposed for HU under the influence of high noise. Besides, [16] is also compared in this article to test the proposed distance metric SSD. It is worth noting that the comparison algorithms are all added a nonnegative constraint to the abundance matrix and endmember matrix on all data, especially to meet this requirement in the iterative process.

D. Parameter Settings

There are four parameters in this article: the spatial window size r, the number of similarity pixels k, the parameter λ_1 for spatial similarity constraint, and the parameter λ_2 for sparse constraint. Here, we use Case 4 and Case 6 datasets to analyze the influence of these four parameters on the unmixing results since the noise interference of them is more complicated. All experiments use the control variable method to determine the final parameters. First, we analyze the effect of spatial window size r on unmixing. According to experience, we first fix λ_1 and λ_2 to 0.01, respectively. The parameter r is defined as {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. The unmixing results are shown in Fig. 3. Analyzing the results of two datasets, we can conclude that, when $r = 4$, SAD and RMSE have the smaller value at the same time. With the increase in r, the errors of both endmember and abundance are increasing. This is because, during the smoothing and unmixing process, the excessively large window introduces a lot of dissimilarity information. Fig. 4 illustrates the effect of k on the experimental results. In fact, according to the unmixing results, k has different effects on each dataset. Here, we set k–10.

Then, we define the sets of λ_1 and λ_2 as {0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5} and then use grid-search to find the best parameters in the sets. Figs. 5 and 6, respectively, give the results of parameter analysis of the algorithm on two data. It can be seen that the performance of the algorithm has a different sensitivity to different parameters on different data, which is also a common problem facing the field of unmixing. After weighing, we set λ_1 and λ_2 to 0.25 and 0.45 in our all experiments, respectively. It is worth noting that, in the specific unmixing, these parameters can be changed correspondingly with the change of the image.

E. Performance Analysis

Table I shows the experimental results of different unmixing methods on six synthetic data. Among them, the bold part indicates the best unmixing results. According to the results, we can conclude as follows.

1) The sparsity of the abundance matrix is a very important property, and the sparsity constraint can promote the algorithm's performance to a large extent.
2) The spatial information can greatly improve unmixing performance. CSNMF, NMF-QMV, SGSNMF, and

[2]https://www.lx.it.pt/ bioucas/code/unmixing_overview.zip
[3]https://github.com/Xinyu-Wang/SGSNMF_TGRS
[4]https://github.com/LinaZhuang/NMF-QMV_demo
[5]https://staffhome.ecm.uwa.edu.au/ 00053650/code.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

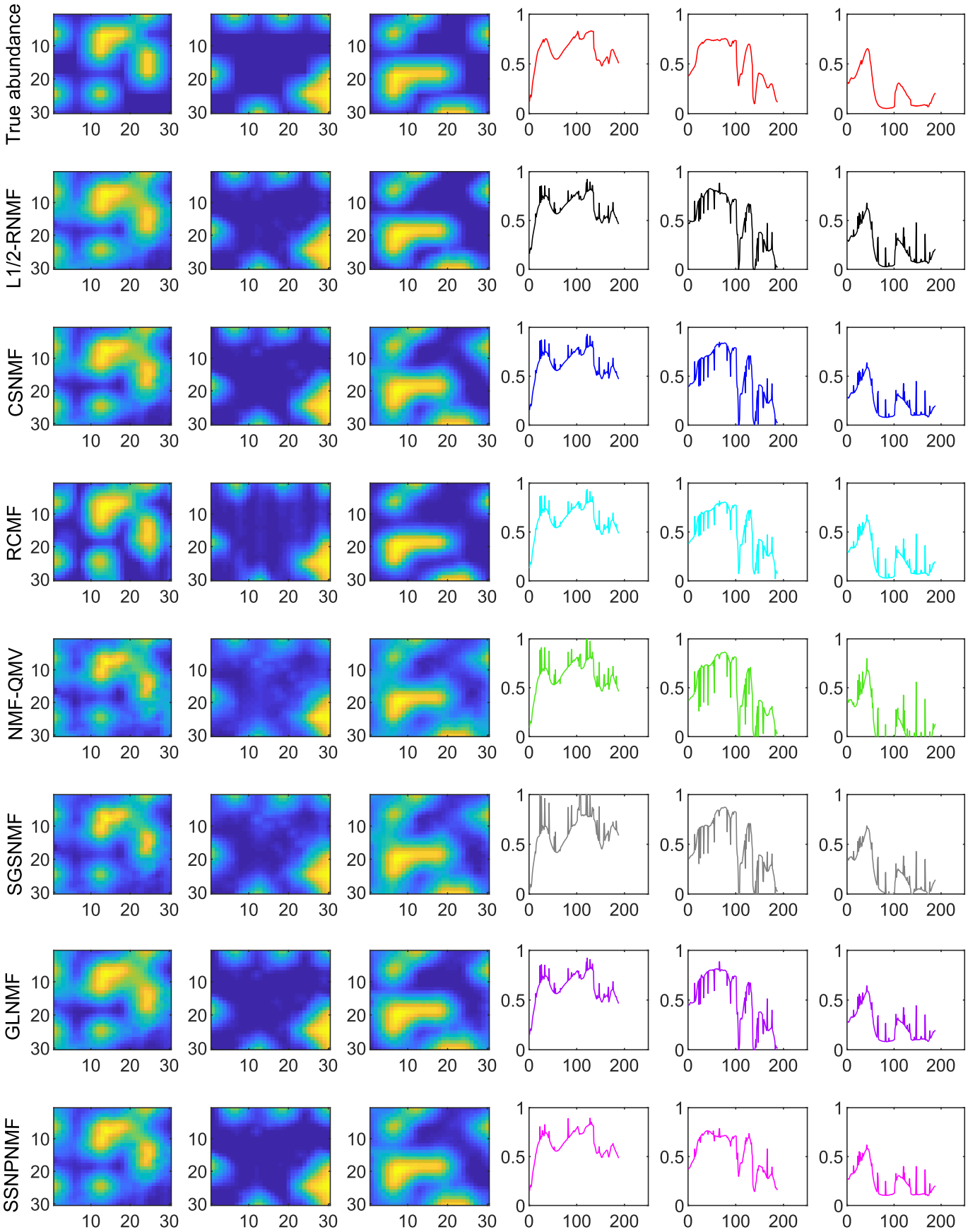


Fig. 7. Example of the unmixing results on Case 6 data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

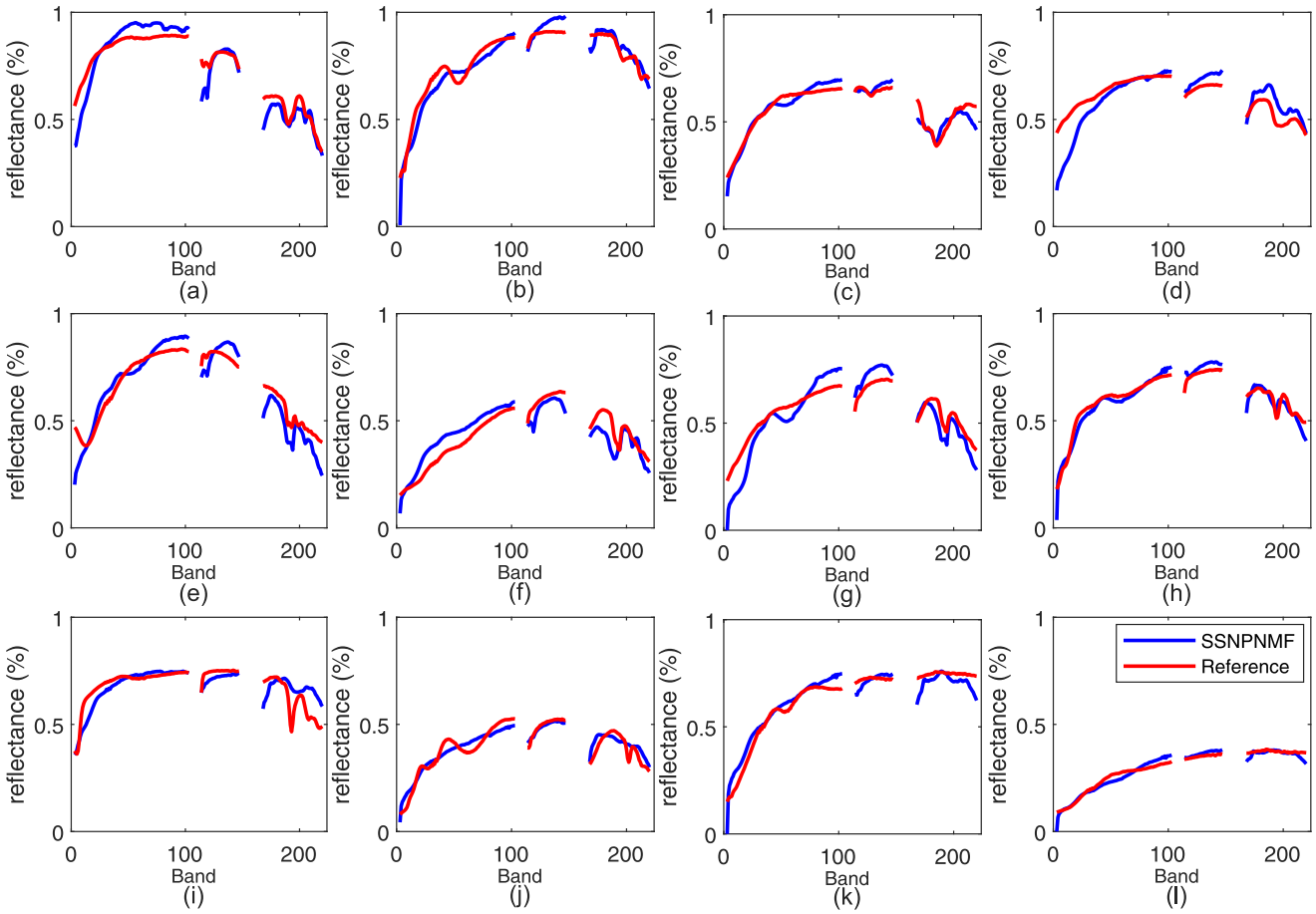


Fig. 8. Comparison of the USGS library spectra (red line) with the endmember signatures extracted by SSNPNMF (blue line) on the AVIRIS Cuprite data. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Chalcedony. (e) Dumortierite. (f) Kaolinite #1. (g) Kaolinite #2. (h) Montmorillonite. (i) Muscovite. (j) Nontronite. (k) Pyrope. (l) Sphene.

our proposed algorithm SSNPNMF all consider spatial information. Compared with VCA-FCLS, RCMF, and $L_{1/2}$-NMF, they can usually achieve better unmixing results. However, for images that are more affected by noise, inaccurate spatial information will make the unmixing performance worse.

3) The method proposed in this article has achieved the best results on all data. Especially on the data seriously affected by noise and banding, superior experimental results are still obtained, which further verifies the effectiveness and robustness of SSNPNMF.

Fig. 7 is an example of unmixing results of the three endmembers and their corresponding abundances of each method on Case 6 data. All the spectral curves are not smooth, which is caused by the noise type of Case 6. The interference of missing pixels and deadlines in all bands in Case 6 is random, and all methods have this phenomenon without considering the smoothing of the spectral dimension during the unmixing. Therefore, it can be taken into account in the future direction of unmixing.

V. Experiments With Real Data

In real experiments, we use three real hyperspectral datasets[6] to verify the effectiveness of the proposed SSNPNMF, which are collected by AVIRIS sensor in the wavelength range of 0.37–2.5 μm. Specifically, the unmixing results of all algorithms on the three data are described in detail in the following.

A. Cuprite Data

The size of Cuprite is $512 \times 512 \times 224$. The coverage area is a mining area, which is widely used as a basic database in the unmixing of HSIs [34], [48], [49]. However, since some known bands are interfered with by strong water vapor, we first remove these bands. Finally, a $200 \times 150 \times 188$ subcube is selected as the experimental data to ensure the effectiveness of all algorithms. According to recent unmixing results [49], there are mainly 12 types of minerals contained in this area: Alunite, Kaolinite #1, Kaolinite #2, Sphene, Andradite, Nontronite, Muscovite, Pyrope, Chalcedony, Dumortierite, Montmorillonite, and Buddingtonite.

The comparison results of all endmember spectra are shown in Fig. 8. Among them, the spectral curves are interrupted because the bands that are severely disturbed by water vapor are removed during data preprocessing. From Fig. 8, it can be seen that the spectra obtained by SSNPNMF are close to the reference spectra. The abundance map of Cuprite data obtained by our algorithm SSNPNMF is shown in Fig. 9. Compared with [50], our abundance map is more accurate. Since

[6]http://lesun.weebly.com/hyperspectral-data-set.html

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SAD AND THE STANDARD DEVIATIONS RESULTS ON THE AVIRIS CUPRITE DATA

	$L_{1/2}$-RNMF	CSNMF	RCMF	NMF-QMV	SGSNMF	GLNMF	SSNPNMF
Alunite	0.1299±0.0681	0.1054±0.0216	0.1027±0.0248	**0.0860**	0.1748±0.0676	0.1170±0.0502	0.1038±0.0646
Andradite	0.0745±0.0210	0.0844±0.0432	0.0835±0.0670	0.1212	**0.0688±0.0021**	0.0731±0.0168	0.0994±0.0286
Buddingtonite	0.0836±0.0196	0.1123±0.0172	0.1054±0.0119	0.0797	0.1094±0.0162	0.0759±0.0062	**0.0699±0.0204**
Chalcedony	0.1430±0.0414	0.0832±0.0541	0.0916±0.0363	0.1428	0.1399±0.0301	0.1438±0.0208	**0.0821±0.0427**
Dumortierite	0.1020±0.0278	0.1390±0.0450	0.0943±0.0725	0.1108	0.1044±0.0100	**0.0778±0.0112**	0.1253±0.0209
Kaolinite#1	0.0935±0.0164	0.1147±0.0916	0.0982±0.1242	**0.0795**	0.1038±0.0159	0.0851±0.0146	0.1226±0.0177
Kaolinite#2	0.0899±0.0280	0.1041±0.1841	0.0872±0.0845	0.1014	0.0878±0.0272	**0.0853±0.0298**	0.0859±0.0219
Montmorillonite	0.0638±0.0205	0.0706±0.0755	0.0814±0.0211	0.1114	0.0811±0.0099	0.0848±0.0388	**0.0628±0.0226**
Muscovite	0.1567±0.0605	**0.0534±0.0085**	0.0951±0.0690	0.0811	0.0841±0.0281	0.1025±0.0359	0.0911±0.0535
Nontronite	0.0913±0.0169	0.1363±0.0407	0.0978±0.0188	0.0715	0.0844±0.0105	0.0902±0.0114	**0.0708±0.0143**
Pyrope	0.1446±0.0642	0.0806±0.0245	0.0914±0.0388	0.1812	0.1046±0.0509	0.1467±0.0056	**0.0565±0.0554**
Sphene	**0.0585±0.0041**	0.0601±0.0590	0.0893±0.0995	0.1879	0.1371±0.0974	0.0596±0.0450	0.0642±0.0541
Mean	0.1132±0.0131	0.0953±0.0715	0.0932±0.0654	0.1190	0.1067±0.0305	0.0951±0.0238	**0.0898±0.0347**

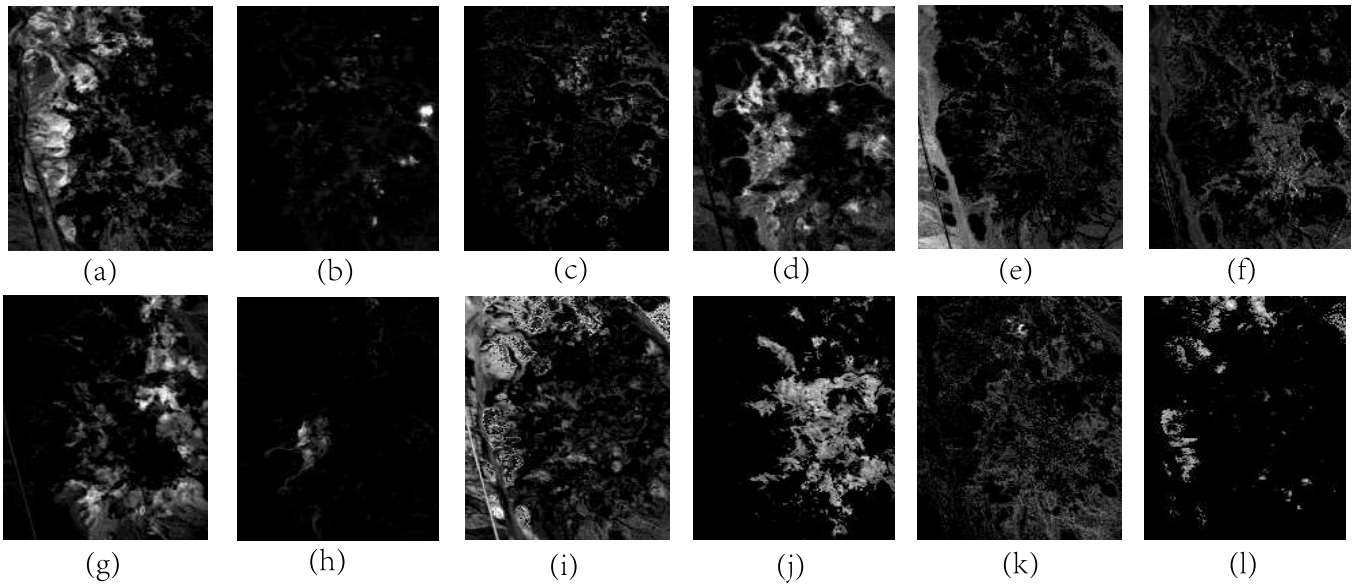


Fig. 9. Abundance maps of different endmembers using SSNPNMF on the AVIRIS Cuprite data. (a) Sphene. (b) Muscovite. (c) Kaolinite #2. (d) Montmorillonite. (e) Andradite. (f) Dumortierite. (g) Alunite. (h) Buddingtonite. (i) Pyrope. (j) Chalcedony. (k) Nontronite. (l) Kaolinite #1.

TABLE III
SAD AND THE STANDARD VARIANCE RESULTS ON THE AVIRIS INDIAN PINES DATA

	$L_{1/2}$-RNMF	CSNMF	RCMF	NMF-QMV	SGSNMF	GLNMF	SSNPNMF
Man-made land	**0.1231±0.0283**	0.2517±0.3817	0.1123±0.1216	0.1651	0.3439±0.2205	0.1232±0.0283	0.1954±0.0136
Wheat	0.2710±0.2455	0.1567±0.0847	**0.0618±0.1444**	0.2629	0.1266±0.0618	0.2701±0.2455	0.0624±0.0161
Corn	0.1266±0.0618	0.1068±0.0984	0.0943±0.0817	0.6889	0.0608±0.0235	0.3251±0.1300	**0.0366±0.0059**
Soybean	0.1130±0.0612	**0.0721±0.0130**	0.0732±0.0209	0.1822	0.6599±0.1947	0.1131±0.0612	0.0910±0.0159
Vegetation	0.0717±0.0162	0.0508±0.1498	0.1427±0.3294	0.0797	0.0270±0.0151	0.0718±0.0162	**0.0436±0.0102**
Haystack	0.0966±0.1322	0.0642±0.1381	0.2308±0.4480	0.0854	0.0611±0.0363	0.0966±0.1322	**0.0266±0.0109**
Mean	0.1289±0.0336	0.1052±0.2145	0.1192±0.1878	0.3209	0.3225±0.0340	0.2289±0.0336	**0.0960±0.0022**

there is no ground truth for abundance, only the quantitative comparison results of endmembers are given here. As shown in Table II, it can be concluded that the SAD and their standard deviations of our unmixing algorithm SSNPNF from the reference on each spectrum are small, there are five optimal values, and the average SAD is also the smallest, reaching 0.0898. SSNPNMF can obtain more accurate results in the unmixing of real scene data since it adopts the calculation mode of spatial–spectral neighborhood preservation, which can more accurately explore the spatial structure of the data and can supplement the spectral information to restore the abundances and endmembers. In summary, compared with the other six popular methods, SSNPNMF has significant advantages in HU.

B. Indian Pines Data

The Indian Pines data are also widely used in denoising tasks due to the large interference from noise [51], [52]. Therefore, we use it here to test the unmixing performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SAD AND THE STANDARD VARIANCE RESULTS ON THE AVIRIS JASPER RIDGE DATA

	$L_{1/2}$-RNMF	CSNMF	RCMF	NMF-QMV	SGSNMF	GLNMF	SSNPNMF
Tree	0.1411±0.0111	0.1480±0.0552	0.1583±0.0010	0.2571	0.1955±0.0013	0.1283±0.0179	**0.1254±0.0032**
Soil	0.4154±0.0731	0.3716±0.0378	**0.3273±0.0023**	1.1890	0.5131±0.0011	0.3860±0.0480	0.4094±0.0216
Water	1.1927±0.2411	1.0474±0.4550	1.2229±0.0097	0.1386	0.1943±0.0069	0.7015±0.4358	**0.1246±0.0615**
Road	**0.1243±0.0222**	0.3218±0.4021	0.1745±0.0037	0.1840	0.2994±0.0063	0.5587±0.4558	0.1483±0.0502
Mean	0.6433±0.0992	0.6520±0.0917	0.4707±0.0042	0.4421	0.3005±0.0039	0.5682±0.1402	**0.2541±0.1328**

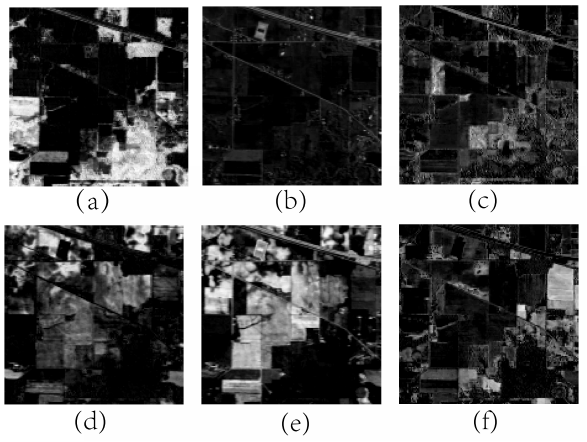


Fig. 10. Abundance maps of different endmembers on the Indian Pines Data. (a) Man-made land. (b) Wheat. (c) Corn. (d) Soybean. (e) Vegetation. (f) Haystack.

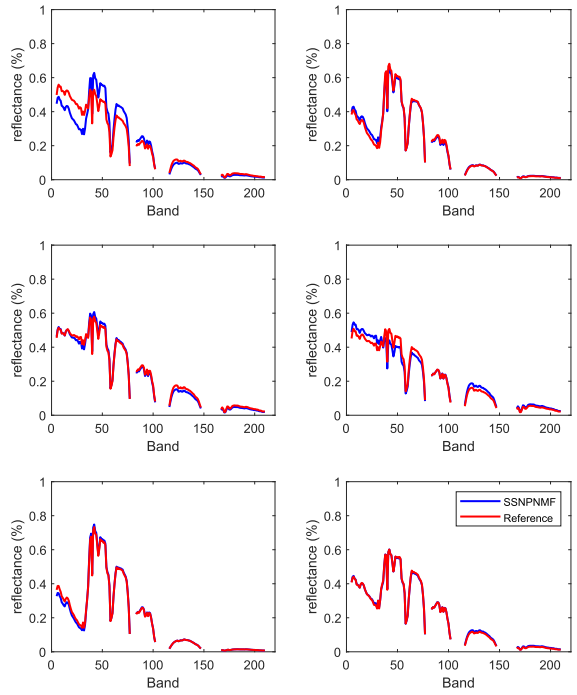


Fig. 11. SSNPNMF extracted and reference spectrum curve comparisons on the Indian Pines Data.

of SSNPNMF in the presence of severe noise. According to Yuan *et al.* [50], after removing the bands with low SNR and water vapor interference, data with the size of $145 \times 145 \times 169$ are retained. According to the known prior, there are six types of endmembers, namely, man-made land, wheat, corn, soybean, vegetation, and haystack.

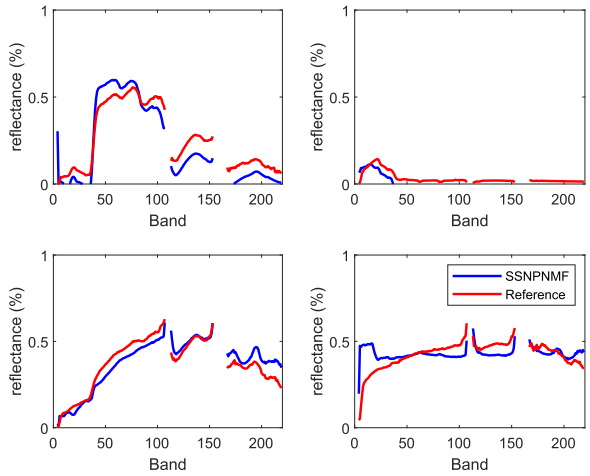


Fig. 12. SSNPNMF extracted and reference spectrum curve comparisons on the Jasper Ridge Data.

The abundance comparison results of all algorithms are presented in Fig. 10. As shown in Fig. 11, it can be seen that the spectral curves of SSNPNMF are closer to the references. Table III shows the SAD comparison results of the seven algorithms. It can be concluded that, except for RCMF and SSNPNMF, the unmixing results of other algorithms in India data are relatively poor. This is because the database itself is greatly affected by noise, and there are a large number of mixed noises, such as the Gaussian noise and the banding noise. These algorithms do not take these factors into account. RCMF has a diagonal matrix specially designed to suppress complex noise in unmixing, which can effectively reduce the influence of noise in the image. In addition, our algorithm designs a distance metric based on space–spectral, which can also effectively improve the unmixing performance.

C. Jasper Ridge Data

Jasper with a size of $100 \times 100 \times 198$ is also used as real data to verify the validity of SSNPNMF. According to Qian *et al.* [53], Jasper contains four materials, namely, tree, soil, water, and road. Fig. 12 shows the comparison results of the four endmember spectra extracted by SSNPNMF and the reference spectrum. The abundance maps of the seven comparison algorithms are shown in Fig. 13. It can be seen that $L_{1/2}$-RNMF, CSNMF, and RCMF are less effective in extracting land and road abundance maps. The abundances extracted by the proposed SSNPNMF are closest to the reference map. Table V shows the SAD comparison results. Table IV shows the RMSE results of all algorithms on the Jasper Ridge Data, which is consistent with the visualization

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

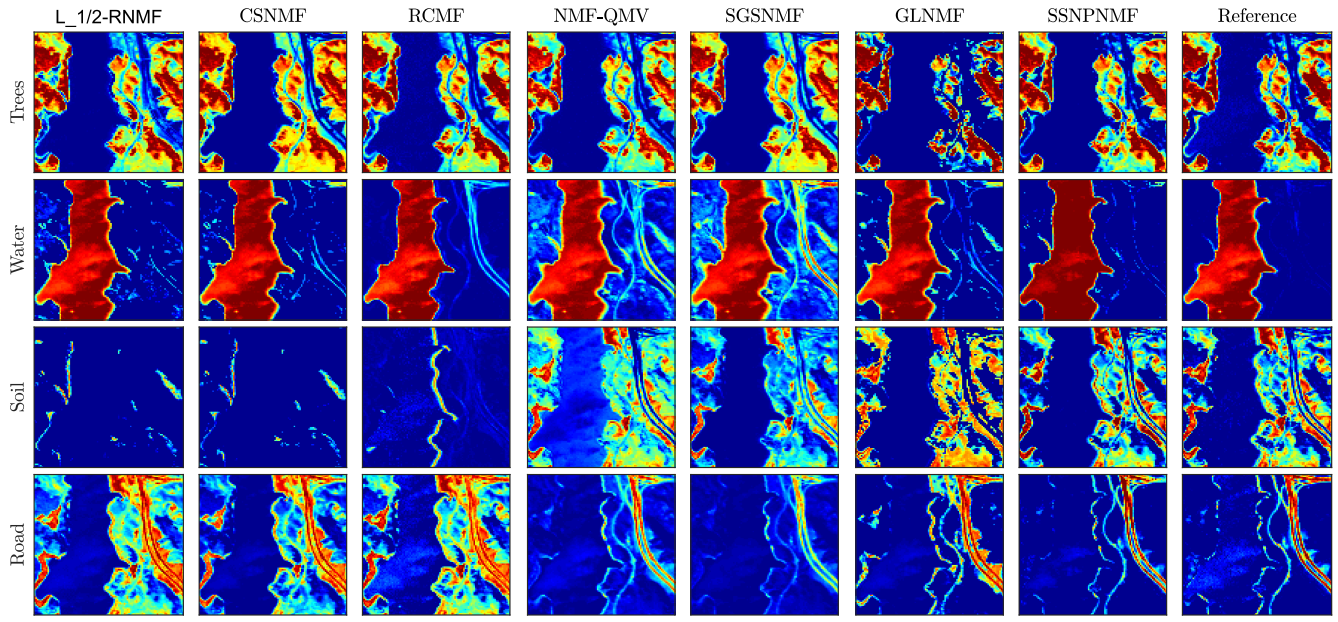


Fig. 13. Abundance maps of different endmembers on the Jasper Ridge Data.

TABLE V
RMSE OF ALL ALGORITHMS ON THE AVIRIS JASPER RIDGE DATA

	$L_{1/2}$-RNMF	CSNMF	RCMF	NMF-QMV	SGSNMF	GLNMF	SSNPNMF
Jasper Ridge Data	0.2653	0.2532	0.2561	0.1356	0.1521	0.1418	0.0638

TABLE VI
RUNNING TIME (SECONDS) OF ALL ALGORITHMS FOR THREE REAL HYPERSPECTRAL DATASETS

	$L_{1/2}$-RNMF	CSNMF	RCMF	NMF-QMV	SGSNMF	GLNMF	SSNPNMF
Cuprite Data	310	1438	227	185	187	26346	96 (20328)
Indian Pines Data	144	803	84	45	104	10346	42 (8524)
Jasper Ridge Data	72	364	37	19	41	1396	25 (747)

result in Fig. 13. It can be concluded that the unmixing results on the Jasper Ridge Date are generally worse than the Cuprite and Indian Data. This may be due to the fact that there are more interactions between the endmembers and the ground objects on Jasper data, and the nonlinear and linear mixed models work together. We will continue to study this issue in future work.

In this section, we also record the time required for all algorithms to run on three real experiments. All experiments are performed on a Linux system with Intel Xeon CPU E5-2697 v3 at 2.60 GHz and 64-GB RAM. As shown in Table VI, two times are provided for the proposed algorithm SSNPNMF, the first one is the time it takes to find the similarity, and the second is the time it takes to converge. It can be seen from Table VI that the NMF-QMV algorithm has the shortest running time on all data. The iterative speed of the proposed SSNPNMF is very fast, but the process of calculating the similarity is very time-consuming. From the time-consuming and performance comparison of all algorithms, it can be understood that the proposed algorithm SSNPNMF consumes more time on very large databases but has a superior performance. It trades time for superior unmixing performance. Therefore, this is also a focus of our follow-up work. We will study how to quickly unmix on the basis of ensuring the performance of the algorithm.

VI. CONCLUSION

In this article, we propose a novel unmixing method called SSNPNMF for incomplete and noisy HSI data, which combines spatial and spectral information simultaneously. SSNPNMF designs a new similarity distance used to select the similarity pixels in the HSI, aiming to adapt to complex data. Then, a graph is created to adjust the reconstruction weight of the pixel and its neighbors according to spatial–spectral information. Subsequently, the reconstruction matrix is used to constrain the abundance to improve unmixing performance. Then, we design six types of complex data, including different degrees of noise, dead pixels, and deadlines. Finally, SSNPNMF is applied to six synthetic datasets and three real data, and the experimental results indicate the effectiveness of our proposed algorithm.

Since NMF needs to reshape 3-D HSI into 2-D images during the unmixing, a lot of useful information for unmixing may be lost in this process. Therefore, we will focus on exploring unmixing methods without reshaping steps in the future.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] A. F. H. Goetz, G. Vane, J. E. Solomon, and B. N. Rock, "Imaging spectrometry for Earth remote sensing," *Science*, vol. 228, no. 4704, pp. 1147–1153, 1985.

[2] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019.

[3] Y. Li, W. Xie, and H. Li, "Hyperspectral image reconstruction by deep convolutional neural network for classification," *Pattern Recognit.*, vol. 63, pp. 371–383, Mar. 2017.

[4] Y. Zhou, H. Chang, K. Barner, P. Spellman, and B. Parvin, "Classification of histology sections via multispectral convolutional sparse coding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 3081–3088.

[5] X.-H. Han, B. Shi, and Y. Zheng, "Self-similarity constrained sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 11, pp. 5625–5637, Nov. 2018.

[6] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.

[7] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[8] G. Liu, Y. Yuan, Y. Zhang, Y. Dong, and X. Li, "Style transformation-based spatial–spectral feature learning for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Oct. 6, 2021, doi: 10.1109/TGRS.2020.3026099.

[9] S. Wang, T.-Z. Huang, X.-L. Zhao, G. Liu, and Y. Cheng, "Double reweighted sparse regression and graph regularization for hyperspectral unmixing," *Remote Sens.*, vol. 10, no. 7, p. 1046, 2018.

[10] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[11] M. Wang, B. Zhang, X. Pan, and S. Yang, "Group low-rank nonnegative matrix factorization with semantic regularizer for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1022–1029, Apr. 2018.

[12] L. Tong, J. Zhou, X. Li, Y. Qian, and Y. Gao, "Region-based structure preserving nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1575–1588, Apr. 2017.

[13] Z. Zhang, S. Liao, H. Zhang, S. Wang, and Y. Wang, "Bilateral filter regularized L2 sparse nonnegative matrix factorization for hyperspectral unmixing," *Remote Sens.*, vol. 10, no. 6, pp. 816–827, 2018.

[14] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[15] E. M. M. B. Ekanayake *et al.*, "Enhanced hyperspectral unmixing via non-negative matrix factorization incorporating the end member independence," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 2256–2259.

[16] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[17] L. Tong, J. Zhou, B. Qian, J. Zhou, and B. Qian, "Adaptive graph regularized multilayer nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 434–447, 2020.

[18] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[19] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[20] Y. E. Salehani and S. Gazor, "Smooth and sparse regularization for NMF hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3677–3692, Aug. 2017.

[21] Z. Guo, T. Wittman, and S. Osher, "L1 unmixing and its application to hyperspectral image enhancement," *Proc. SPIE*, vol. 7334, Apr. 2009, Art. no. 73341M.

[22] Z. B. Xu, H. Zhang, Y. Wang, X. Y. Chang, and Y. Liang, "$L_{1/2}$ regularization," *Sci. China Inf. Sci.*, vol. 53, no. 6, pp. 1159–1169, 2010.

[23] Z. Xu, X. Chang, F. Xu, and H. Zhang, "$L_{1/2}$ regularization: A thresholding representation theory and a fast solver," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 7, pp. 1013–1027, Jul. 2012.

[24] W. Wang and Y. Qian, "Adaptive $L_{1/2}$ sparsity-constrained NMF with half-thresholding algorithm for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2618–2631, Jun. 2015.

[25] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Hyperspectral unmixing with l_q regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6793–6806, Nov. 2014.

[26] W. He, H. Zhang, and L. Zhang, "Sparsity-regularized robust non-negative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4267–4279, Sep. 2016.

[27] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[28] C. Li, X. Chen, and Y. Jiang, "On diverse noises in hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5388–5402, Oct. 2015.

[29] X.-R. Feng, H.-C. Li, J. Li, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using sparsity-constrained deep nonnegative matrix factorization with total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6245–6257, Oct. 2018.

[30] B. Rathnayake, E. M. M. B. Ekanayake, K. Weerakoon, G. M. R. I. Godaliyadda, M. P. B. Ekanayake, and H. M. V. R. Herath, "Graph-based blind hyperspectral unmixing via nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6391–6409, Sep. 2020.

[31] A. Erturk, "Coupled nonnegative matrix factorization with local neighborhood weights for data fusion," in *Proc. Medit. Middle-East Geosci. Remote Sens. Symp. (MGARSS)*, Mar. 2020, pp. 41–44.

[32] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

[33] Q. Gu and J. Zhou, "Neighborhood preserving nonnegative matrix factorization," in *Proc. BMVC*, 2009, pp. 1–10.

[34] X. Lu, L. Dong, and Y. Yuan, "Subspace clustering constrained sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3007–3019, May 2020.

[35] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Jun. 2011.

[36] W. Wang, Y. Qian, and Y. Y. Tang, "Hypergraph-regularized sparse NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 681–694, Feb. 2016.

[37] G. Cui, X. Li, and Y. Dong, "Subspace clustering guided convex nonnegative matrix factorization," *Neurocomputing*, vol. 292, pp. 38–48, May 2018.

[38] H. Huang, G. Shi, H. He, Y. Duan, and F. Luo, "Dimensionality reduction of hyperspectral imagery based on spatial–spectral manifold learning," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2604–2616, Jun. 2020.

[39] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. ICCV*, Jan. 1998, vol. 98, no. 1, pp. 839–846.

[40] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[41] D. Cai, X. He, J. Han, and T. S. Huang, "Graph regularized nonnegative matrix factorization for data representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1548–1560, Aug. 2011.

[42] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[43] R. N. Clark, G. A. Swayze, T. V. King, A. J. Gallagher, and W. M. Calvin, "The U.S. Geological Survey, Digital Spectral Reflectance Library: Version 1: 0.2 to 3.0 microns," U.S. Geol. Surv., Open File Rep. 93-592, 1993.

[44] C. Cao, J. Yu, C. Zhou, K. Hu, F. Xiao, and X. Gao, "Hyperspectral image denoising via subspace-based nonlocal low-rank and sparse factorization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 3, pp. 973–988, Mar. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.

[45] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.
[46] X. Lu, H. Wu, and Y. Yuan, "Double constrained NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2746–2758, May 2014.
[47] L. Zhuang, C. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.
[48] N. Akhtar and A. Mian, "RCMF: Robust constrained matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3354–3366, Jun. 2017.
[49] Y. Yuan, Y. Feng, and X. Lu, "Projection-based NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2632–2643, Jun. 2015.
[50] Y. Yuan, M. Fu, and X. Lu, "Substance dependence constrained sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 2975–2986, Jun. 2015.
[51] Y. Zheng, T. Huang, X. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020.
[52] Y. Yuan, H. Ma, and G. Liu, "Partial-DNet: A novel blind denoising model with noise intensity estimation for HSI," *IEEE Trans. Geosci. Remote Sens.*, early access, May 3, 2021, doi: 10.1109/TGRS.2021.3071799.
[53] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

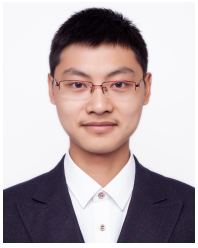


Le Dong is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China.

Her is currently working as an Exchange Student with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China. Her main research interests are hyperspectral image analysis, pattern recognition, and machine learning.



Xiaoqiang Lu (Senior Member, IEEE) is currently a Full Professor with Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, Shaanxi, China, and also with the University of Chinese Academy of Sciences, Beijing, China. His research interests include pattern recognition, machine learning, hyperspectral image analysis, cellular automata, and medical imaging.



Ganchao Liu received the Ph.D. degree from Xidian University, Xi'an, China, in 2016.

He is currently an Associate Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an. His interests include image processing and pattern recognition.

Yuan Yuan (Senior Member, IEEE) is currently a Full Professor with the School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an, China. Her research interests include visual information processing and image/video content analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:28:38 UTC from IEEE Xplore. Restrictions apply.